



Lesson 4: Classification and Evaluation Metrics

Technologies for Artificial Intelligence

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Before we start

- Exercise 3, which we discuss now
- Ridge, Lasso, and the choice of the penalty strength

Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**

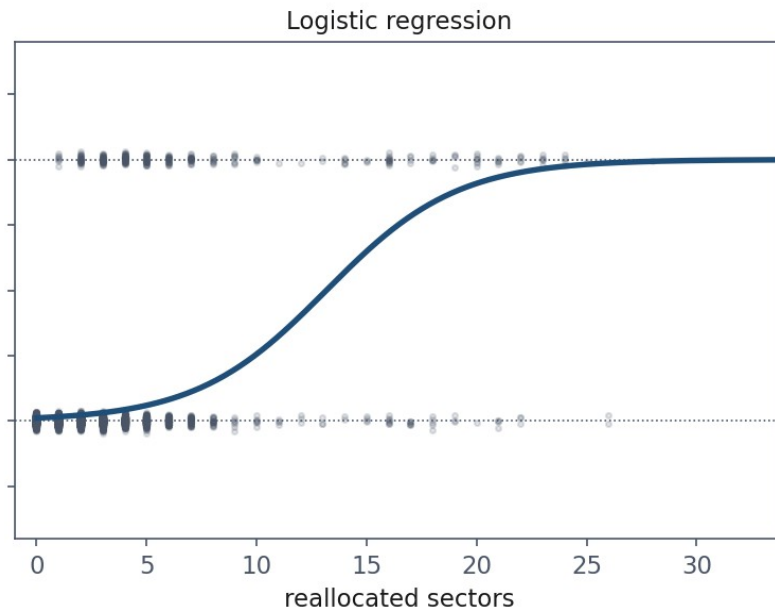
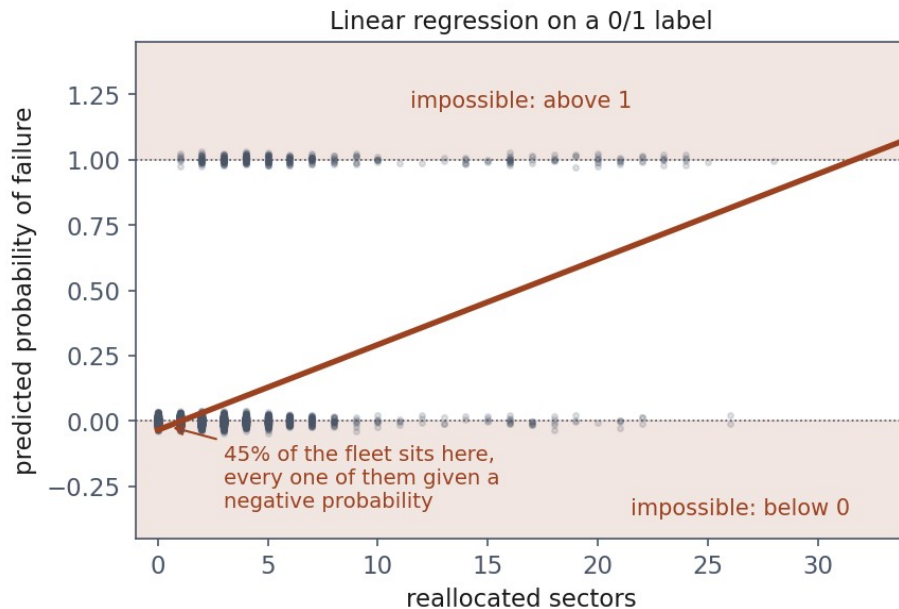
The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Label: did it fail within thirty days?

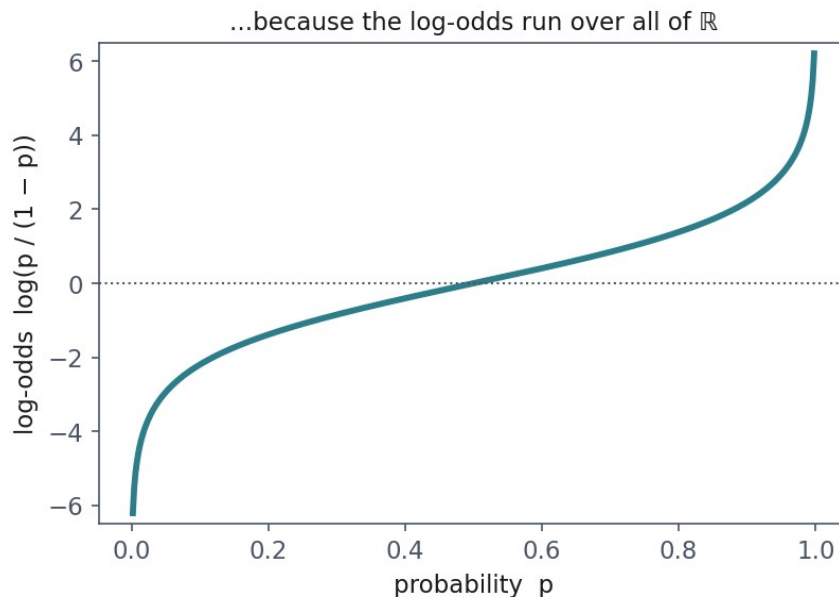
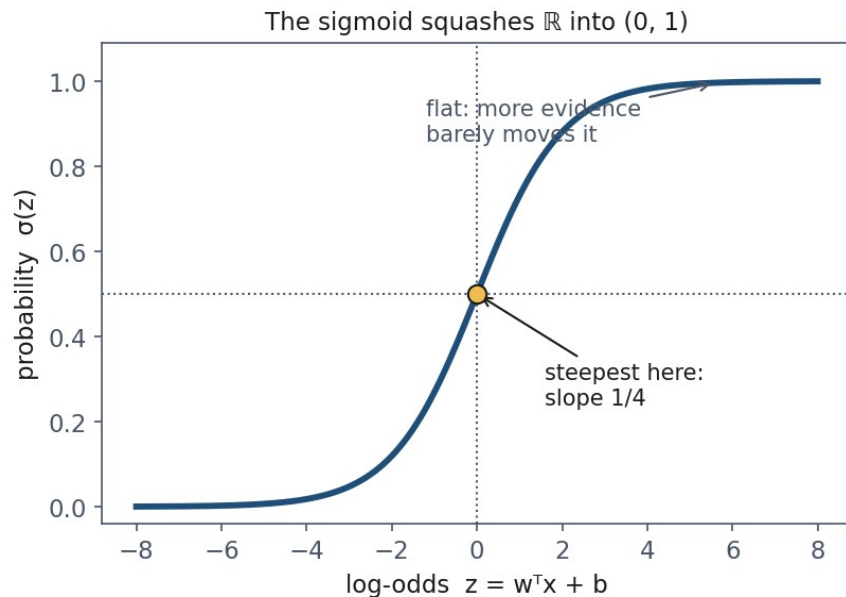
306

- **306 of the 8,000 drives failed.** 3.8% of the fleet
- Which means 96.2% did not

What goes wrong if we just fit a line



The sigmoid



The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

What the coefficients mean

- The model is linear in the log-odds, so $\exp(\text{coefficient})$ multiplies the **odds**
- Features standardised, so one unit = one standard deviation
- A coefficient of 1.80 means **six times the odds** of failing

The truth, and what we recovered

Feature	True	Fitted	Odds ×
reallocated_sectors	1.80	1.66	5.28
spin_retry_count	1.15	1.17	3.22
read_error_rate	0.85	0.87	2.39
temperature_c	0.45	0.47	1.59
power_on_hours	0.35	0.38	1.46
seek_error_rate	0.00	0.02	1.03

The intercept is the base rate

- Fitted intercept: **-6.09**
- $\sigma(-6.09) = 0.0023$: a 0.2% chance for an average drive

Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid's output?

Maximum likelihood, in one sentence

- The model has a knob; each setting assigns a probability to what happened
- **Turn the knob so that what actually happened is least surprising**

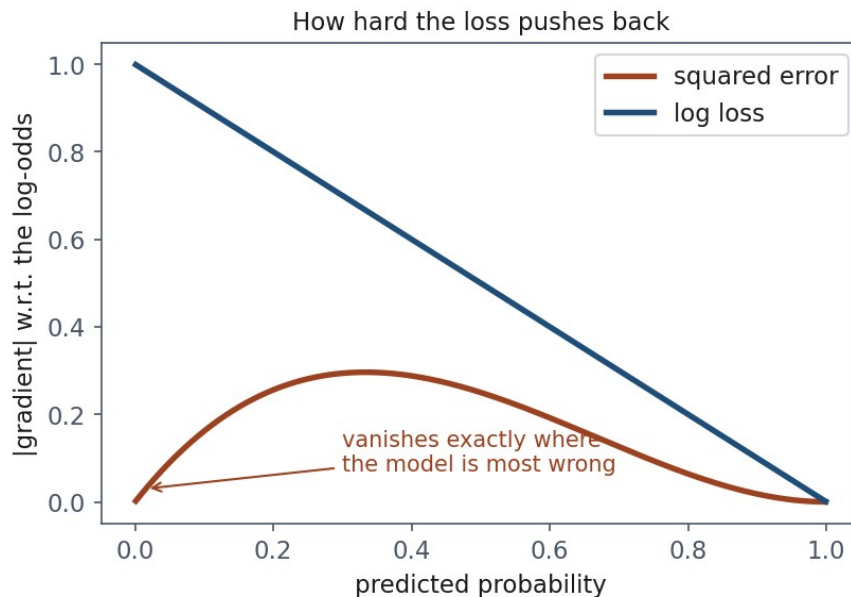
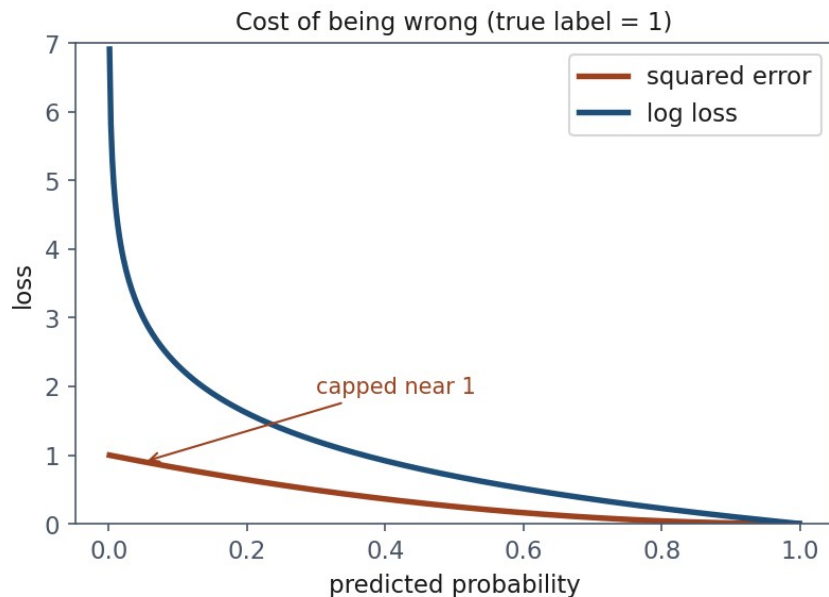
Cross-entropy, also called log loss

$$L = - [y \log p + (1 - y) \log(1 - p)]$$

The gradient is the one you already know

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

So why not squared error?



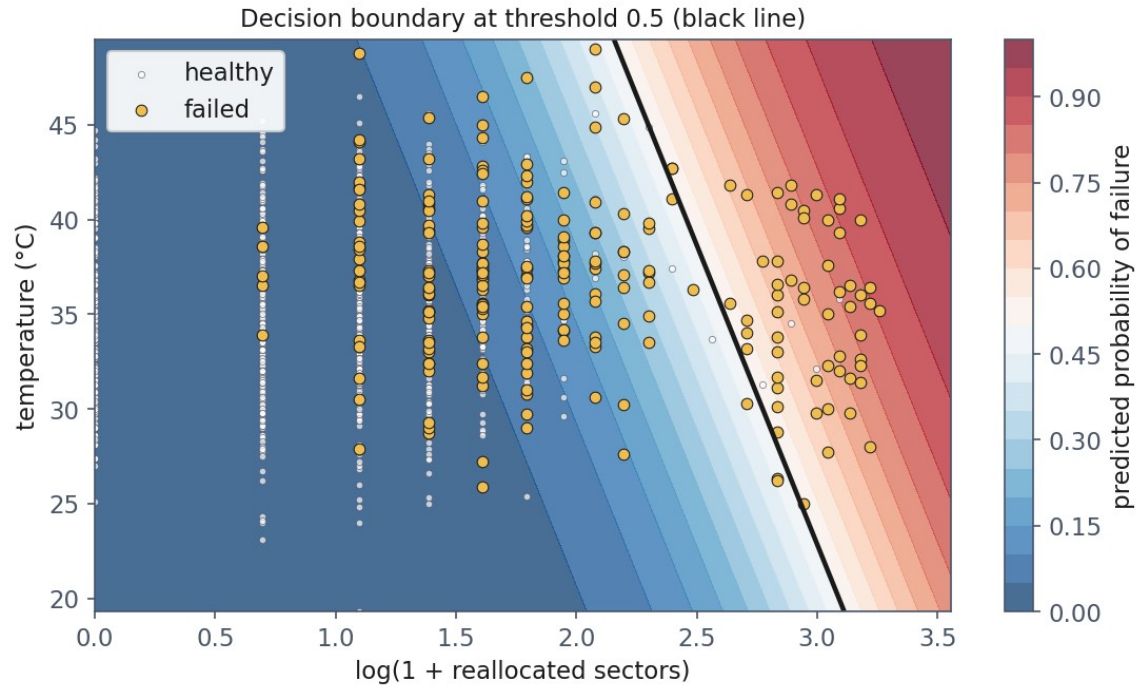
And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not

Notebook 1, live

- Fifteen lines of NumPy, then one line of scikit-learn
- Same answer

The decision boundary



Break

- Ten minutes

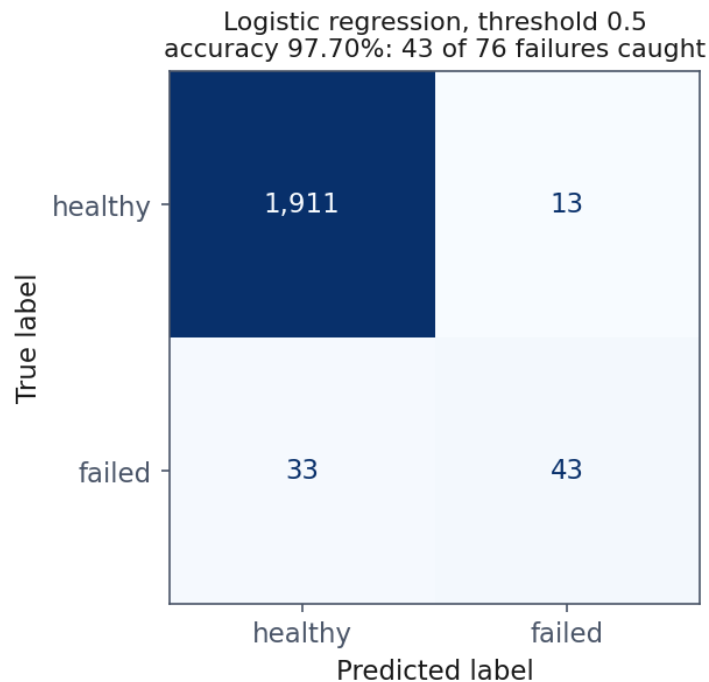
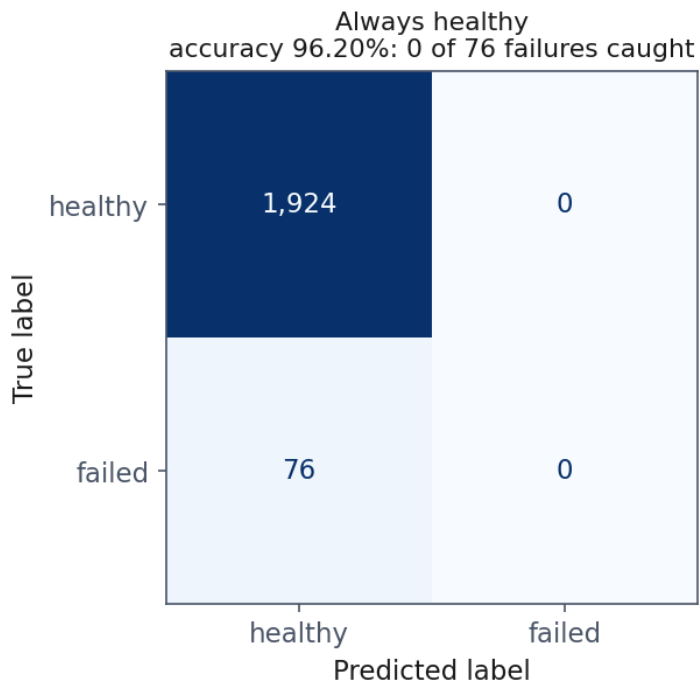
Two models

Model	Accuracy	Failures caught
Predict “healthy” for everything	96.20%	0 of 76
Logistic regression, threshold 0.5	97.70%	43 of 76

Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem that is exactly what you want
- The interesting class is almost always the **rare** one

Four numbers instead of one



The four outcomes, and what they cost

- **False positive:** we replace a healthy drive. A technician's hour, €140
- **False negative:** a drive dies in service. Callout, rebuild, risk, €2,600

Precision and recall: look at the denominator

Precision = $TP / (TP + FP)$
“of the alarms we raised”

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	FN 33	TP 43

Recall = $TP / (TP + FN)$
“of the drives that failed”

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	FN 33	TP 43

Our model, in those terms

- Precision **0.768**: of 56 drives flagged, 43 really failed
- Recall **0.566**: of 76 drives that failed, we found 43
- Specificity 0.993: of 1,924 healthy drives, we left 1,911 alone

F1: why the harmonic mean

Model	Precision	Recall	Arithmetic	Harmonic
Flag every drive	0.038	1.000	0.519	0.073

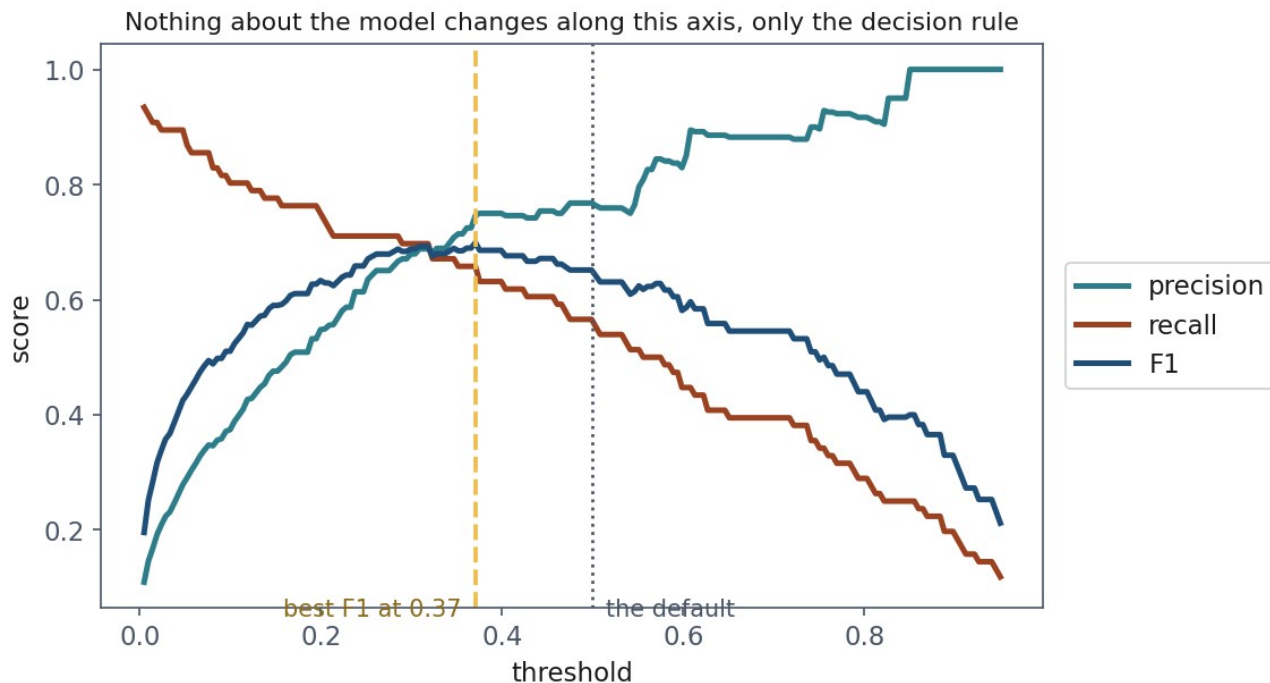
Read the minority class, not the average

- weighted avg F1: **0.975** (dominated by the healthy class)
- macro avg F1: **0.820** (both classes get an equal say)

Notebook 2, live

- The confusion matrix by hand, then from scikit-learn
- Then the threshold sweep

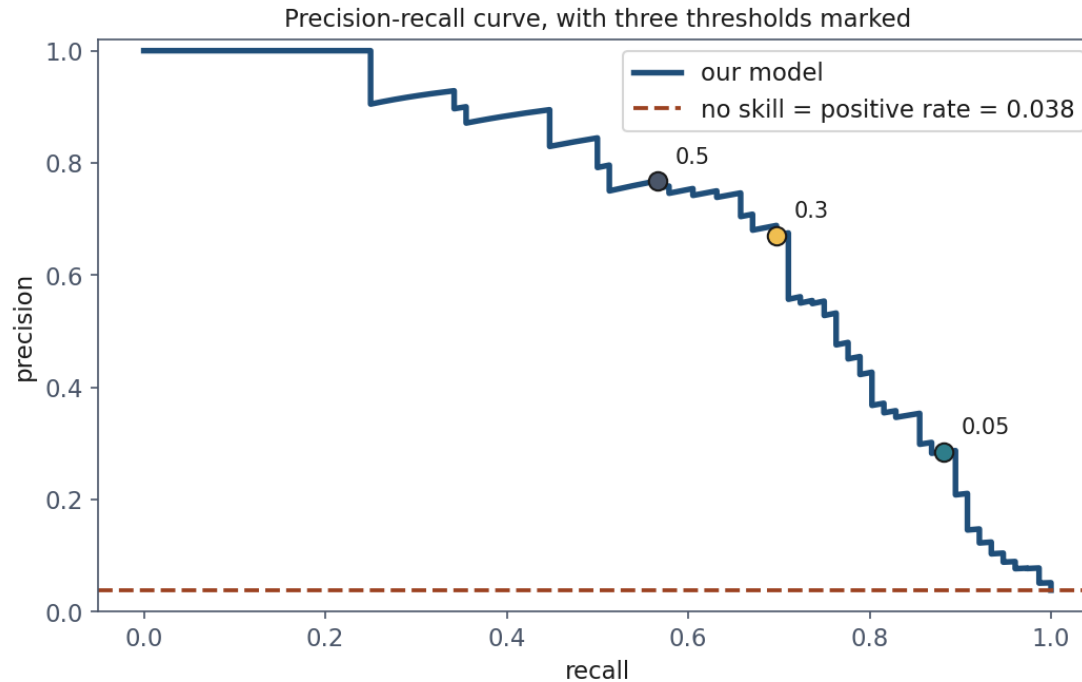
The threshold is a choice nobody made



What the sweep actually costs

Threshold	TP	FP	FN	Precision	Recall	Accuracy
0.02	69	284	7	0.195	0.908	0.855
0.10	61	102	15	0.374	0.803	0.942
0.30	53	25	23	0.679	0.697	0.976
0.50	43	13	33	0.768	0.566	0.977
0.90	14	0	62	1.000	0.184	0.969

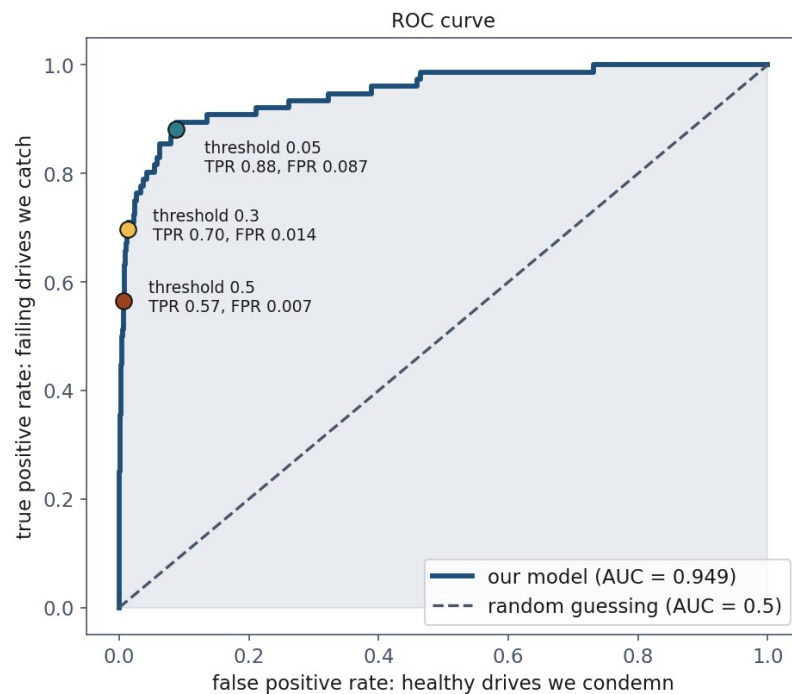
The precision-recall curve



Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- What if we score the whole **ranking** instead?

The ROC curve



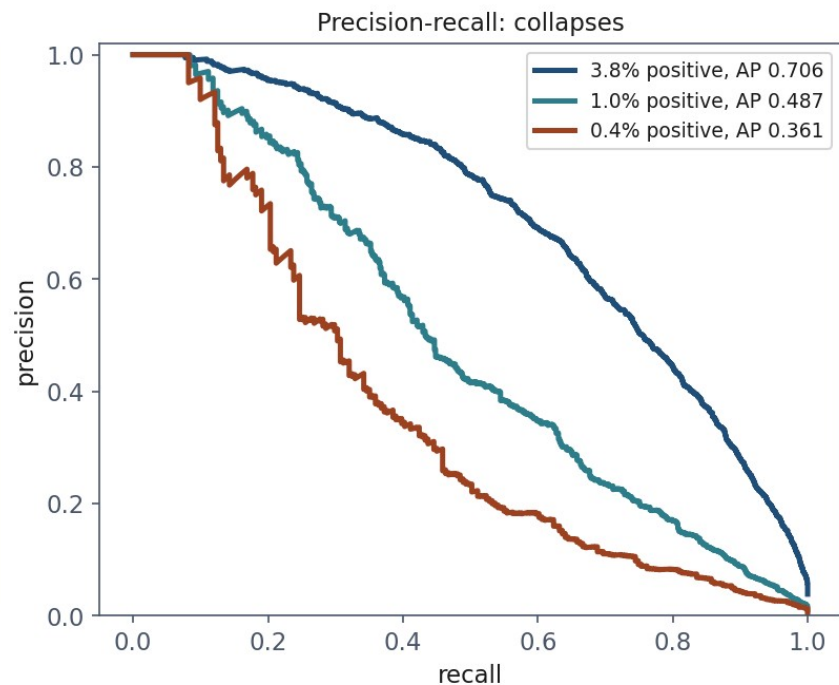
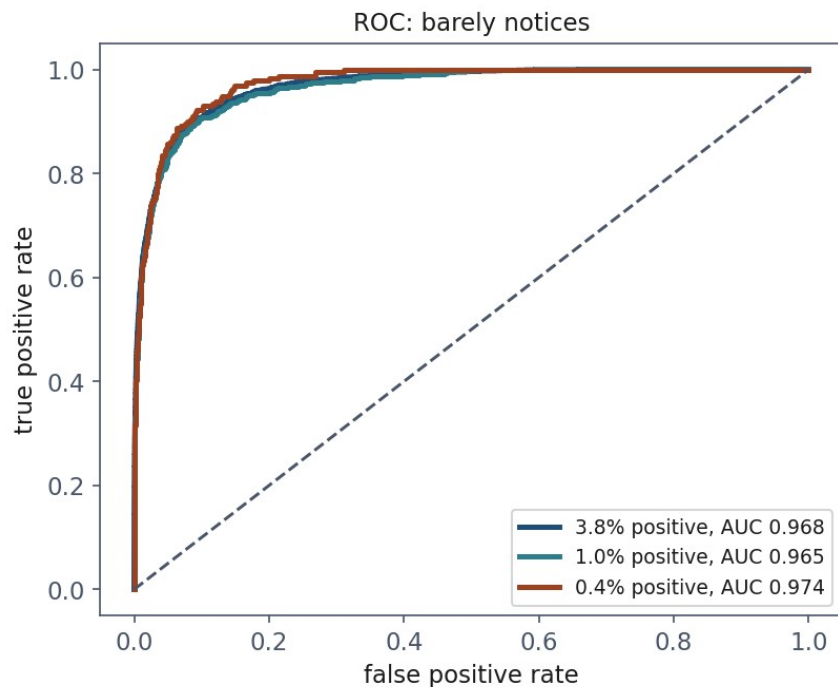
What the area means

- **AUC = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949

What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**; every decision changes

Where ROC misleads



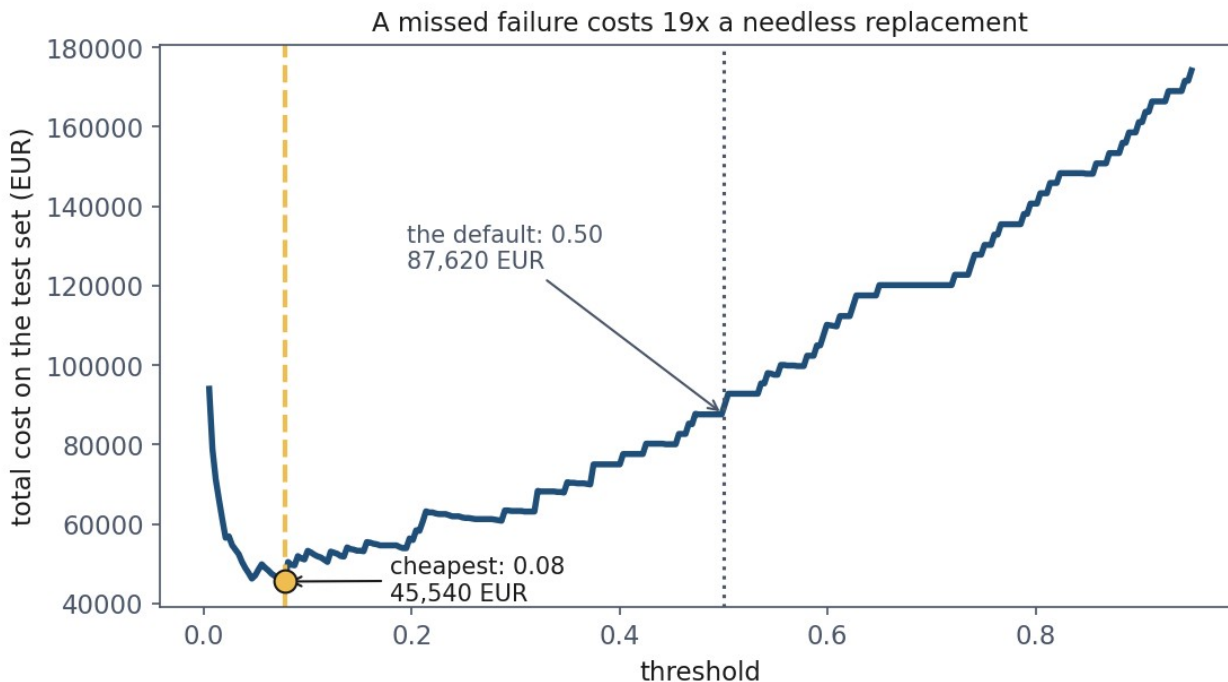
Why the two disagree

- FPR divides by **all healthy drives**: an enormous, growing denominator
- Precision divides by **the alarms raised**, where the false ones dominate

The rule

- On an imbalanced problem, report the **precision-recall curve alongside AUC**
- Never instead of it, and never AUC alone

Choosing the threshold from money



And it has a closed form

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{140 + 2600} = 0.051$$

The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade

Class weights: the same lever, pulled earlier

Model	Recall	Precision	AUC	Cost
Plain, threshold 0.50	0.566	0.768	0.949	87,620
Plain, threshold 0.08	0.855	0.349	0.949	45,540
Balanced weights, 0.50	0.895	0.249	0.950	49,500

Notebook 3, live

- ROC by hand, the 200,000-pair experiment, the cost curve
- Then the multiclass matrix

More than two classes



Macro against weighted, one last time

- macro avg F1: **0.707**
- weighted avg F1: **0.883**

What to take away

- **96.20% accuracy, nothing caught.** Accuracy measures the majority class
- Precision is about the **alarms**, recall is about the **failures**
- The **threshold is a decision**, and it has a formula
- **AUC measures ranking** and hides imbalance

Homework

- **Exercise 4**, discussed at the start of **Friday 23 October**
- Build a classifier, choose a threshold from a stated cost ratio, defend it